

APPLICATIONS

WINTER PEAK ELECTRICITY LOAD FORECASTING IN SOUTH AFRICA USING EXTREME VALUE THEORY

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Key words: Extreme value theory, seasonal peaks, temperature, probability of exceedance.

Summary: The paper explores the use of extreme value theory in calculating the probability of exceedance (POE) levels and posterior predictive densities of peak electricity demand in South Africa for the period 2000 to 2009. In this paper the exceedances above a sufficiently large threshold are modelled through a Generalized Pareto-type (GP-type) distribution. The GP-type distribution is derived as an approximation of the tail of a Generalized Burr-Gamma (GBG) distribution (a convenient extreme value distribution fitted to the whole data set) by Verster and De Waal (2011). The main advantage of this modelling approach is that the GP-type distribution has only one parameter to estimate and not two as is the case with the well-known Generalized Pareto

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distribution (GPD). An application of a Generalized Pareto-type (GP-type) in modelling daily peak electricity demand is presented. The optimum threshold is determined using a Generalized Pareto quantile plot. Empirical results show that the GP-type distribution is appropriate in modelling of extreme daily peak electricity demand above a sufficiently high threshold. A comparative analysis was done by fitting a Generalized Extreme Value (GEV) distribution. Empirical results show that the Weibull class of distributions can be used to model the daily peak electricity demand data. The results from the GP-type distribution are better than those from fitting a GEV distribution.

1. Introduction

Predictions of abnormal peak loads are important to load forecasters who must provide accurate predictions to system operators in the electricity sector. The peak forecasts assist the system operators in planning and scheduling of electricity. In order to evaluate peak demand forecasts, it is important to analyse forecast peak demand distributions. A forecast distribution consists of all possible probability of exceedance (POE) values. A forecast distribution gives possible values which future demand could take on. Modelling of exceedances above a sufficiently high threshold has been studied for over three decades. Research has shown that the distribution function of the excesses above a high threshold converges to a generalized Pareto distribution (GPD) as the threshold tends to the right endpoint (Balkema and de Haan, 1974; Pickands, 1975).

If we let $X_1, X_2, \dots, X_n$ to be a sequence of hourly electricity demand, then the upper extremes from this sequence can be extracted by taking the exceedances over a predetermined high threshold u . The exceedances y_j over u are those x_j such that $x_j > u$. The values $y_j = x_j - u$ are the excesses over

u . Then if $F^{[u]}(b_u + a_u x)$ has a continuous limiting distribution function as u goes to the right endpoint, $\omega(F)$, then

$$\left| F^{[u]}(x) - W_{\gamma, \mu, \sigma}(x) \right| \rightarrow 0, \text{ as } u \rightarrow \omega(F) \quad (1)$$

for some Generalized Pareto distribution function, $W_{\gamma, \mu, \sigma}(x)$, with shape, location and scale parameters γ , μ and σ respectively and for a suitable choice of constants $b_u > 0$ and $a_u > 0$ (Balkema and de Haan, 1974; Pickands, 1975). $F^{[u]}$ represents the empirical exceedance distribution function at u .

Recent work includes that of Verster and De Waal (2011) who showed that the tail of a Generalized Burr-Gamma (GBG) distribution can be approximated by a Generalized Pareto-type (GP-type) distribution. The GBG class of distributions is popular in extreme value theory when the whole data set is modelled (Verster and De Waal, 2011). When only considering the observations above a threshold they have derived a Generalized Pareto-type (GP-type) distribution to approximate the tail of the GBG. This modelling approach is convenient since the GP-type distribution has only one parameter to estimate given that the threshold is known, compared to the two parameters of the usual GPD. In this paper we outline the use of extreme value theory in modelling the probabilities of unusually high electricity demand which is caused by extreme events. These events could be extreme weather and/or non-weather factors. Work in this area includes that of Hor et al. (2008) who assessed load forecast uncertainty using extreme value theory. They concluded that by using a GEV model, the number of return observations exceeding given extreme observations is closely in line with what was predicted. Allen and Morzurch (1995) compared probability forecasts derived from theoretical distributions. A detailed discussion on winter peak demand forecasting is found in (Hyndman and Fan, 2009).

The rest of the paper is organized as follows; in Section 2 we discuss the winter demand characteristics. The GP-type distribution is introduced in Section 3. This section also discusses a Bayesian approach for estimating the unknown parameter of the GP-type distribution as well as the prediction of future tail probabilities through the posterior predictive survival function. Section 4 presents empirical results. The conclusion is covered in Section 5.

2. Winter Demand Characteristics and Demand Data

In this paper, winter demand forecasts have been developed using historical data for 10 winters, from 2000 to 2009. The data is from Eskom, South Africa's power utility company. The winter is defined as the 93 day period from 15 May to 15 August each year. Winter peak demand is the highest level of hourly demand observed during a particular winter. The daily peak demand (DPD) is the maximum hourly demand in a 24-hour period. A daily demand profile is shown in Figure 1.

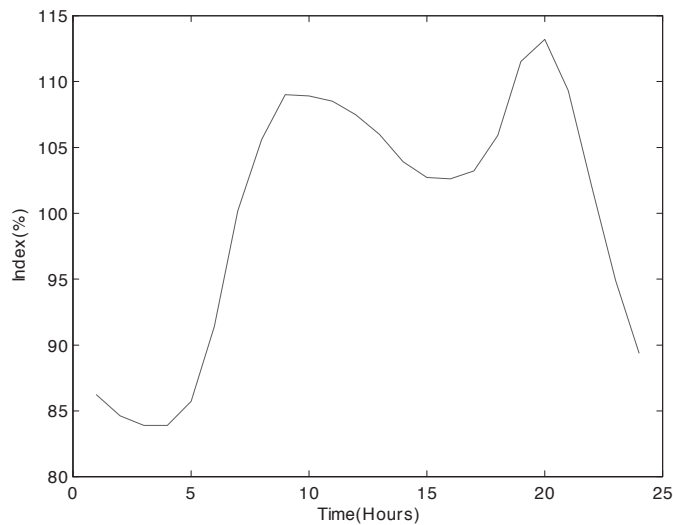


Figure 1. Daily demand profile.

The daily peak is in the evening around 20:00hrs. This data is net energy sent out (NESO) from the distribution division of Eskom in response to some demand of electrical power. NESO (measured in megawatts) is defined as the rate at which electrical energy is delivered to customers. In this paper NESO is used as a proxy of electrical demand after adjusting for energy losses. This definition of electrical demand has its weaknesses. Electrical demand is bounded by the power plants available to provide supply at any time of the day including the need for reserve capacity. Demand cannot exceed supply and there are no market forces acting to influence electricity prices and hence reducing demand in the short run. Prices are generally fixed. If demand were to exceed supply, intervention takes place in the form of for example, load shedding. Load shedding is the last resort used to prevent a system-wide blackout. This NESO definition excludes the demand from people, companies etc. who are willing (or unwilling) and able to pay (or are unable to pay) for electricity but currently do not have access to electrical power. Despite the weakness in the NESO definition of electrical demand, it is still a good and measurable proxy for electrical demand.

Electricity demand in South Africa is highly sensitive to temperature changes in winter and less sensitive in summer (Sigauke and Chikobvu, 2010). The monthly electricity demand profile in Figure 2 shows that the highest

annual peak is in July which is a winter month in the Southern Hemisphere.

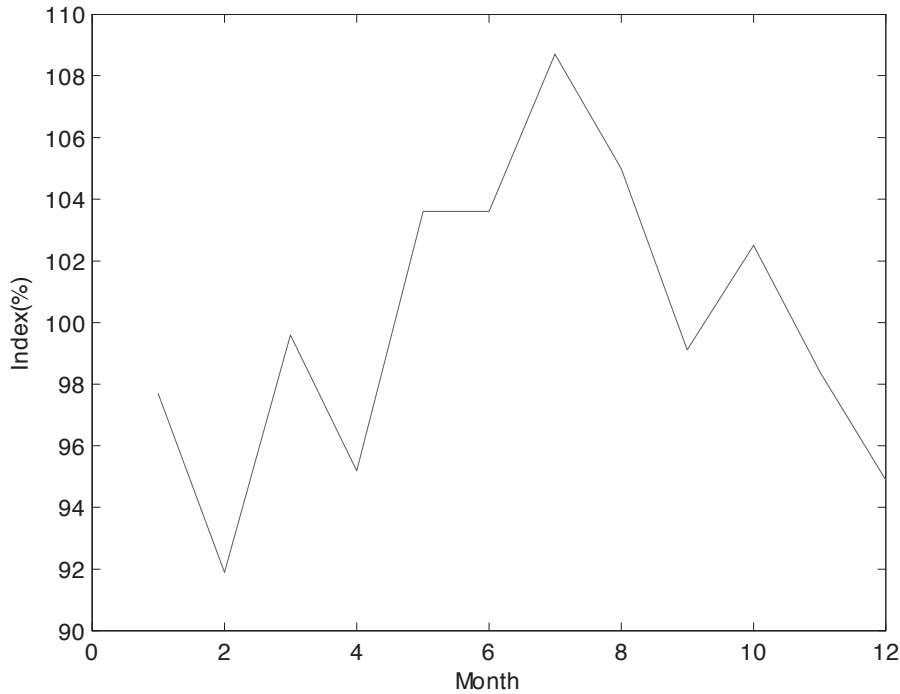


Figure 2. Monthly demand profile.

3. Bayes Estimation of the approximated GP-type Distribution

In Verster and De Waal (2011), it is given that above a reasonably high threshold, t , the tail of a GBG can be approximated by a GP-type distribution with the following survival function:

$$P(V > v | V > t) = \left(1 + \frac{\xi(v - t)}{1 + \xi t}\right)^{-\frac{1}{\xi}}, \quad v > t, \xi > 0 \quad (2)$$

where V represents daily peak electricity demand and $V - t$ is the excess above a high threshold t . The parameter ξ is known as the extreme value index (EVI).

The posterior distribution is given by

$$p(\xi|v, t) \propto p(\xi)p(v|\xi, t) = \frac{e^{-\xi}}{1 + \xi t} \prod_{i=1}^n \frac{1}{1 + \xi t} \left[1 + \frac{\xi(v_i - t)}{1 + \xi t} \right]^{-\frac{1}{\xi}} \quad (3)$$

where $p(v|\xi, t)$ is the likelihood function, $p(\xi) = \frac{e^{-\xi}}{1 + \xi t}$ is the maximum data information prior distribution and n is the number of exceedances above the threshold. See Verster and De Waal (2011) for more detail.

The posterior predictive survival function for predicting the future tail probability, V_0 , is given by

$$\begin{aligned} P(V_0 > v_0|v, t) &\propto \int p(\xi|v, t) \left(1 + \frac{\xi(v_0 - t)}{1 + \xi t} \right)^{-\frac{1}{\xi}} d\xi \\ &= E_{\xi|v} \left(1 + \frac{\xi(v_0 - t)}{1 + \xi t} \right)^{-\frac{1}{\xi}}, -\infty < \xi < \infty. \end{aligned} \quad (4)$$

The latter cannot be computed analytically, but can be approximated easily by simulation.

4. Empirical Results

4.1 The data set

The data used is daily peak electricity demand (DPD) obtained from Eskom, South Africa's power utility company. Aggregated DPD data for the residential, commercial and industrial sectors of the economy is used for the period 2000 to 2009 giving a total of 3636 observations. Electricity demand is generally subject to seasonal changes and an upward positive linear trend. The data is initially transformed by taking natural logarithms. An initial threshold is set at zero after detrending and all those observations with negative values are excluded. The optimum threshold of these observations is then determined using a generalized Pareto quantile plot shown in Figure 3. The threshold is chosen at the observation value where the plot begins to follow a linear

pattern, which is $t = \exp(8) = 2980.96$. There are 81 exceedances above this threshold of $t = 2980.96$.

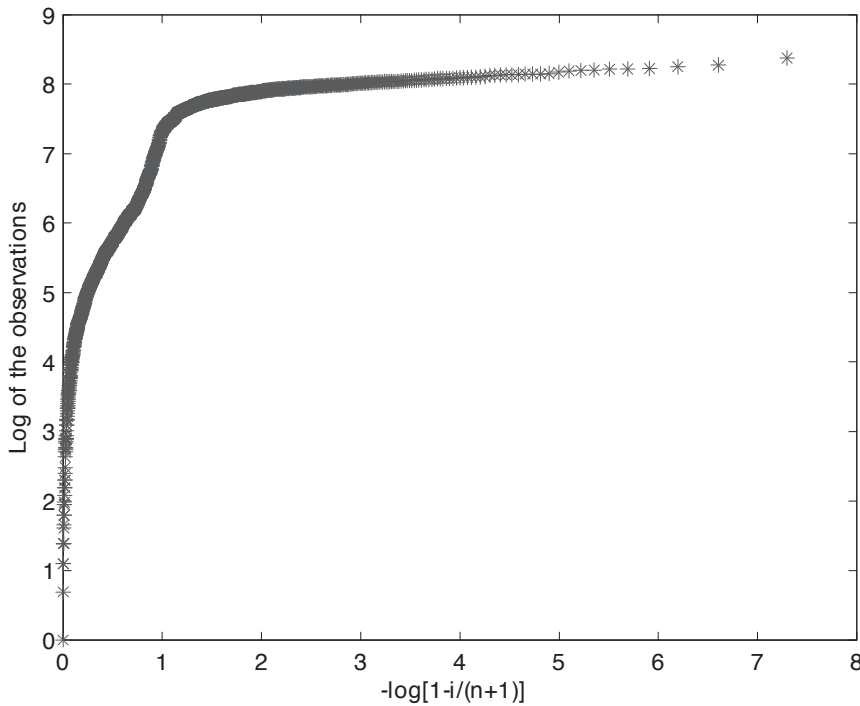


Figure 3. Generalized Pareto quantile plot on the observations greater than zeros.

4.2 Predicting Future Observations

We now consider the observations larger than the threshold t to be GP-type distributed. The parameter ξ is estimated through a Bayesian approach. The posterior distribution of ξ is given in equation (3). The mode of the posterior distribution is taken as the estimate of ξ which is found to be 0.077. The empirical cumulative distribution function (cdf) and the cdf of the GP-type distribution with the estimated parameter value are shown in Figure 4. It can be seen that the fit is close to the observed distribution and this shows that

the GP-type fits well to the distribution of the exceedances. From Figure 4 it seems as if the GP-type is an appropriate model to consider modelling the observations above the threshold.

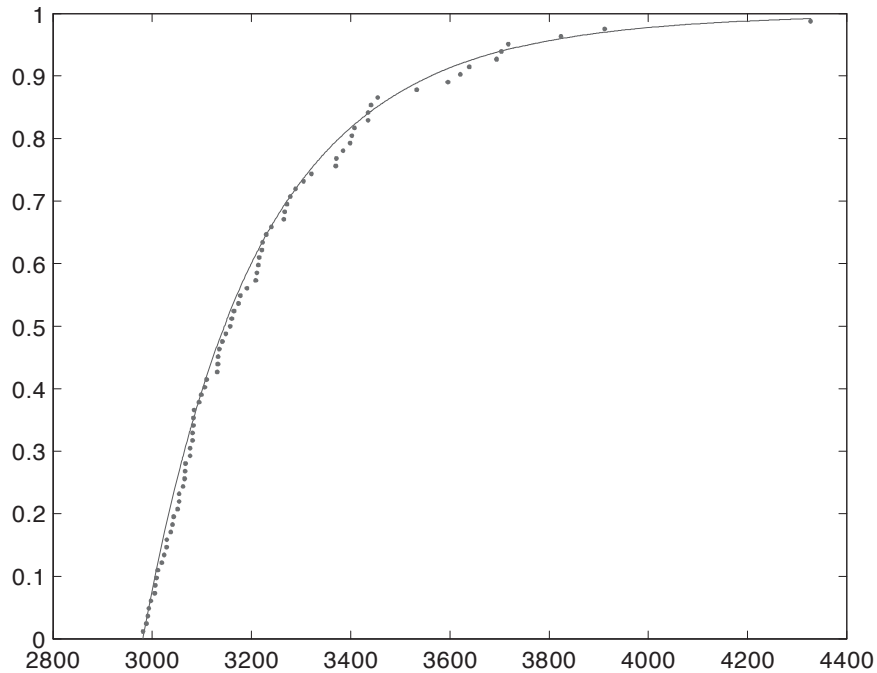


Figure 4. Graphical plot of the empirical *cdf* (dotted curve) and the *cdf* of the GP-type (solid curve) on the exceedances.

The Quantile-Quantile (QQ) plot for the observations above the threshold is constructed in Figure 5. The QQ-plot is a method for judging the goodness of fit of the proposed model to the data, see for example (Beirlant et al., 2004). If the QQ-plot follows a 45° line it indicates a good fit. Figure 5 therefore

indicates a fairly good fit of the GP-type distribution to the data.

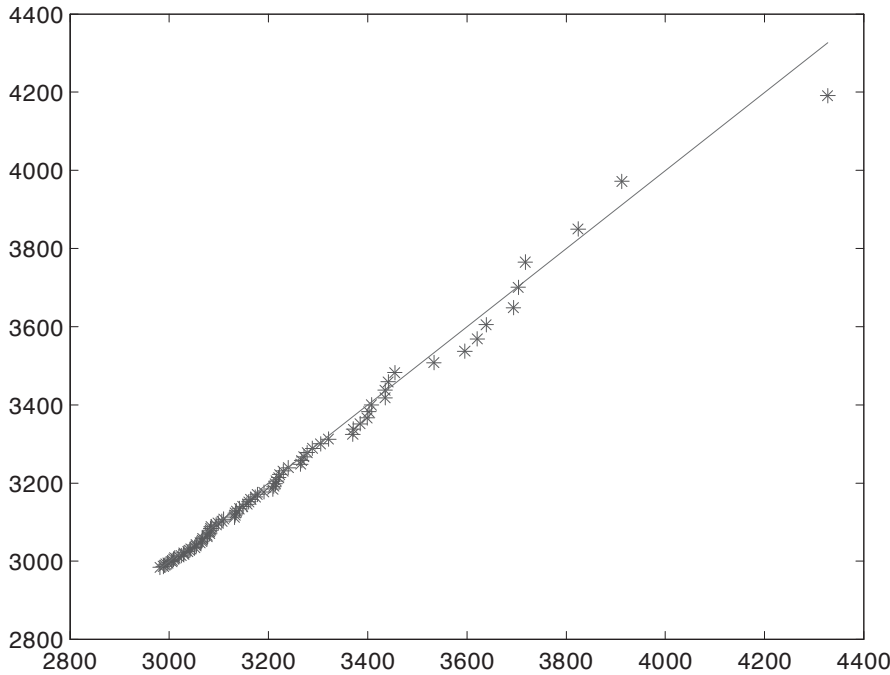


Figure 5. QQ plot of DPD above $t = 2981$. The horizontal axis represents the standard theoretical quantiles while the empirical quantiles are plotted on the vertical axis.

The posterior predictive survival distribution of the approximated GP-type distribution given in equation (4) is now used to predict posterior tail probabilities. For example we can predict the probability exceeding a large value such as 4000, which is

$$P(X_0 > 4000 | x_1, \dots, x_n) = \frac{1}{m} \sum_{j=1}^m \left(1 + \frac{\xi_j(x_0 - t)}{1 + \xi_j t} \right) = 0.0237 \quad (5)$$

where m number of ξ values were simulated from the posterior and substituted into equation (5). ξ_j denotes the value on the j -th simulation run. Some of the predicted tail probabilities for various future daily peak electricity demands

where ξ was simulated from its posterior are given in Table 1 and the density function of the posterior predictive tail probabilities is given in Figure 6.

Table 1 Posterior predictive tail probabilities

x_{n+1}	$P(X > x_{n+1} X > t)$ Posterior predictive tail probabilities
3000	0.9224
3500	0.1310
4000	0.0237
4500	0.0052

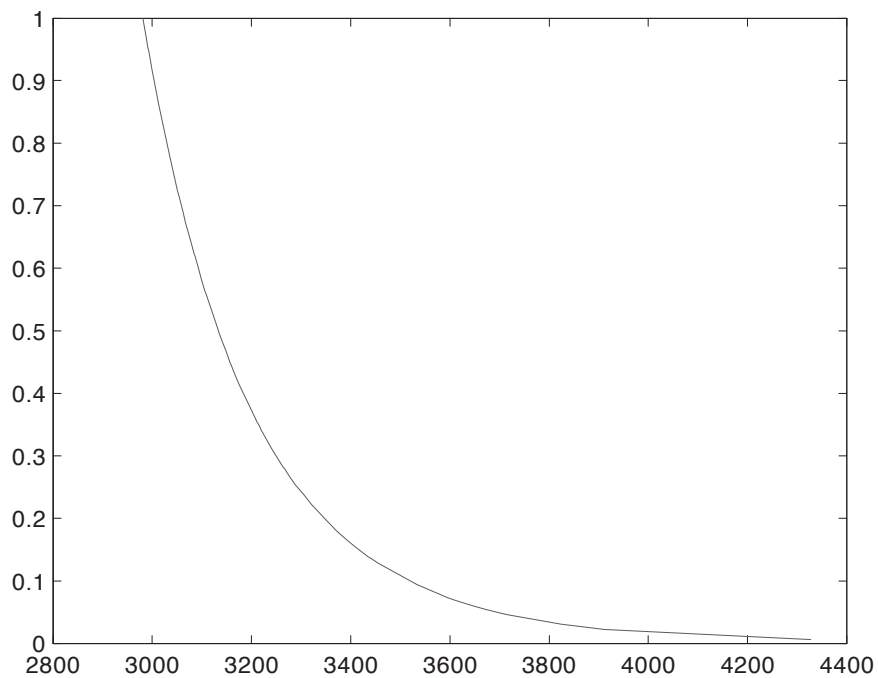


Figure 6. Density function for the posterior predictive tail probabilities.

4.3 Predicting Future Observations Using the GEV distribution: A comparative analysis

In this section we consider fitting a generalized extreme value (GEV) distribution to all the observations (not only the observations above a threshold).

Let $X_1, X_2, \dots, X_n$ be a sequence of hourly electricity demand. Then $M_n = \max(X_1, X_2, \dots, X_n)$ is the maximum hourly electricity demand over an n - observation period where $n \geq 2$. We divide our hourly data into b blocks. Each block consists of 24 hours each, which is a day. The maximum of n observations in block i is then defined as $\{M_{in} = \max(X_{i1}, X_{i2}, \dots, X_{in})\}_i, i = 1, 2, \dots, b$ and $n \geq 2$, with $X_{i1}, X_{i2}, \dots, X_{in}$ the data in the i -th block. In order to extract upper extreme values an extreme value distribution is then fitted to the block maxima data $\{M_{in}\}_i, i = 1, \dots, b$. The probability of M_{in} not exceeding a value x_i is then given as

$$\begin{aligned} P\{M_{in} \leq x_i\} &= P\{\max X_{in} \leq x_i\} \\ &= P\{X_{i1} \leq x_i, X_{i2} \leq x_i, \dots, X_{in} \leq x_i\} \\ &= \{F(x_i)\}^{in}. \end{aligned} \quad (6)$$

The (GEV) distribution consists of the Gumbel, Frechet and Weibull classes which are also known as the type I, II and III extreme value classes respectively. The unified model is given by

$$G_{\gamma, \mu, \sigma}(x) = \exp \left\{ - \left[1 + \gamma \left(\frac{x - \mu}{\sigma} \right) \right]^{\frac{1}{\gamma}} \right\}, 1 + \gamma \left(\frac{x - \mu}{\sigma} \right) > 0, \gamma \neq 0 \quad (7)$$

where γ is the extreme value index. For $\gamma = 0$ equation (7) is the Gumbel class of distributions which is given by

$$G_{\gamma,\mu,\sigma}(x) = \exp \left\{ -\exp - \left[\frac{x - \mu}{\sigma} \right] \right\}, \quad \text{if } \gamma = 0. \quad (8)$$

When $\gamma > 0$ we have the Frechet class of distributions and when $\gamma < 0$ we have the Weibull class of distributions. The three parameters γ , μ , and σ are estimated for the given dataset through the maximum likelihood method. The statistical package MATLAB is used for obtaining the maximum likelihood estimates. The estimates are given as:

$$\gamma = -0.0401, \mu = -540.482, \sigma = 1008.22$$

These results show that the data can be modelled using a Weibull class, ($\gamma < 0$). We then use the quantile function for the unified GEV for predicting the probability of exceedance levels. The quantile function is given by

$$G_{\hat{\gamma},\hat{\mu},\hat{\sigma}}^{-1}(x) = \hat{\mu} - \frac{\hat{\sigma}}{\hat{\gamma}} \left[1 - \{ -\ln(1 - p_i) \}^{-\hat{\gamma}} \right], \hat{\gamma} \neq 0, p_i = \frac{i}{n+1}, i = 1, \dots, n. \quad (9)$$

Some of the predicted tail probabilities for various future daily peak electricity demand using equation (9) are given in Table 2. In Figure 7 the QQ-plot for the theoretical quantiles against the empirical quantiles is constructed for all the observations.

Table 2 Posterior predictive tail probabilities

x_{n+1}	$P(X > x_{n+1} X > t)$ Posterior predictive tail probabilities
3000	0.0225
3500	0.0126
4000	0.0069
4500	0.0038

The results from the GBG approximation to a GP-type distribution are better than those from fitting a GEV distribution as shown by the QQ plots of Figures 5 and 7 and the predictive tail probabilities shown in Tables 1 and 2.

5. Conclusion

This paper has discussed an application of a Generalized Pareto-type (GP-type) distribution to the statistical analysis of extreme daily peak electricity demand. The GP-type distribution seems appropriate in modelling of extreme daily peak electricity demand. The main advantage of this modelling approach is that the GP-type distribution has only one parameter to estimate, given that the threshold is known. A comparative analysis is done by fitting a GEV distribution. Empirical results show that the Weibull class can be used to model the daily peak electricity demand data. The results from the GP-type distribution are better than those from fitting a GEV distribution.

Finally, we state some remaining issues. In this study we did not use the mean excess plot to determine an optimum threshold. It would be interesting to see the nature of the results we get from using this approach. Future research should also look at a comparative analysis of the GPD with the GP-type. Another interesting area of study would involve the use of robust Bayesian

analysis in which a class of prior distributions is used instead of a single prior distribution. These areas will be studied in our future research.

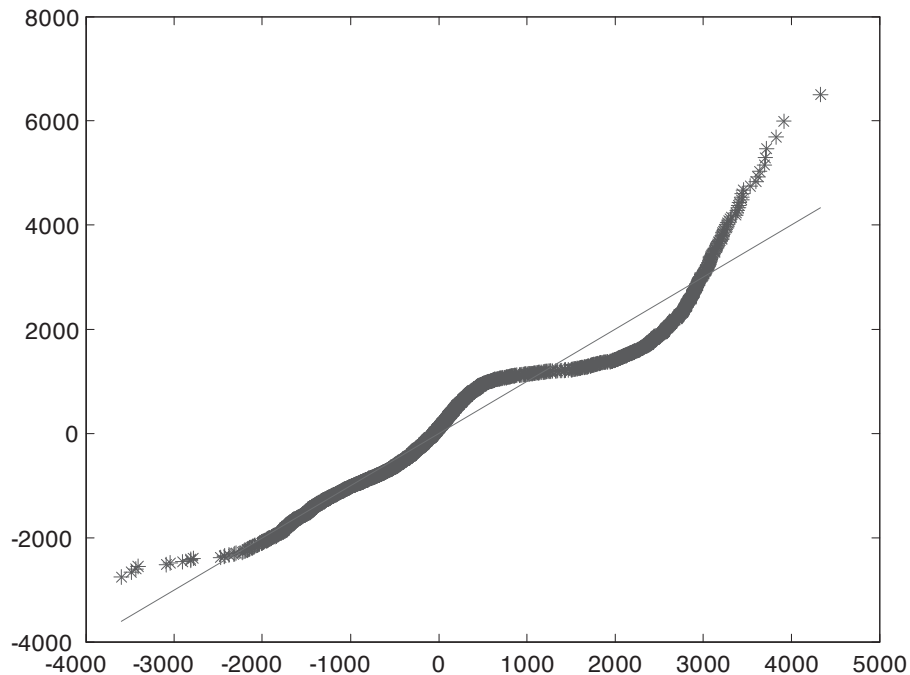


Figure 7. QQ plot of all the DPD observations. The horizontal axis represents the standard theoretical quantiles while the empirical quantiles are plotted on the vertical axis.

Acknowledgements

The authors are grateful to Eskom for providing the data, Department of Statistics and Operations Research, University of Limpopo, Turfloop Campus and Department of Mathematical Statistics and Actuarial Science, University of the Free State for using their resources and to the numerous people who assisted in making comments on this paper.

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